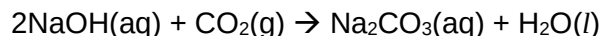


ACJC 2019 H2 Chem Promo Exam Qn Paper

Question 1: To determine the concentration of NaOH and Na₂CO₃ in FA1

Aqueous NaOH can be used to remove CO₂ from exhaled air in spacecrafts. NaOH reacts with CO₂ according to the following equation:



A sample of aqueous NaOH was used to absorb CO₂ in the air for some time. The resulting mixture of NaOH and Na₂CO₃ formed was labelled as **FA1**.

You are to determine the concentration of NaOH and Na₂CO₃ in **FA1** by titration with an aqueous solution of HCl, **FA2**.

You are provided with the following:

FA1, a mixture of NaOH and Na₂CO₃

FA2, 2.50 mol dm⁻³ of hydrochloric acid, HCl

methyl orange indicator

distilled water

(a) Titration with methyl orange as indicator

1. Pipette 25.0 cm³ of **FA1** into a 250 cm³ conical flask and add 1 – 2 drops of methyl orange indicator.
2. Titrate the contents of the flask with **FA2** from the burette until the solution turns orange.
3. Repeat the titration as many times as necessary to obtain consistent results.
4. Record the titration results in the table in part (b). Ensure that the recorded results show the precision of your practical work.

(b) Recording of Results

Titration results with methyl orange

	1	2	
Final burette reading / cm ³	20.20	20.20	
Initial burette reading / cm ³	0.00	0.00	
Volume of FA2 / cm ³			

[3]

Obtain from your titration results a suitable average titre. Place ticks below the readings used to calculate this average titre.

[1]

- (c) A student conducted the titration using 25.0 cm³ of **FA1** against **FA2** using thymol blue as an indicator and obtained the following results.

Titration results with thymol blue

	1	2	3
Final burette reading / cm ³	14.50	28.60	42.75
Initial burette reading / cm ³	0.00	14.40	28.60
Volume of FA2 / cm ³			
Chosen values			

Select appropriate titration values from the table above by placing ticks below the chosen volumes of **FA2**.

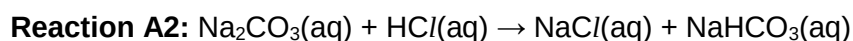
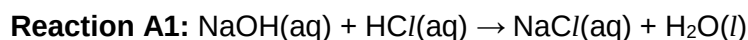
[1]

Use your chosen values to determine the volume of **FA2** reacted with 25.0 cm³ of **FA1**. Show your working below.

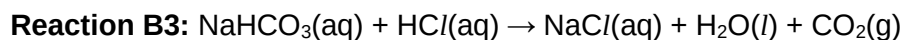
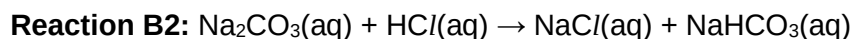
[1]

(d) Calculation

When **thymol blue** is used as the indicator in part (c), the following reactions occur.

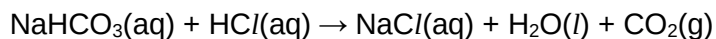


When **methyl orange** is used as the indicator in part (b), the following reactions occur.



The NaHCO₃ in **Reaction B3** is the product from reaction of Na₂CO₃ with HCl in **Reaction B2**.

- (i) Calculate the volume of hydrochloric acid that reacted with 25.0 cm³ of **FA 1** in **Reaction B3**.



[1]

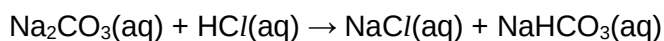
- (ii) Considering **Reactions B2** and **B3**, determine the volume of hydrochloric acid reacted with sodium carbonate in 25.0 cm³ of **FA 1** to form NaHCO₃.

[1]

- (iii) Calculate the volume of hydrochloric acid that reacted with the sodium hydroxide present in 25.0 cm³ of **FA 1**.

[1]

- (iv) Using your answer in **(d)(ii)** and the equation for **Reaction A2**, calculate the amount of sodium carbonate present in 25.0 cm³ of **FA 1**.



[1]

- (v) Calculate the concentration of sodium carbonate, Na₂CO₃ in **FA1**.

[1]

- (vi) From your answer to (d)(iii) and the equation for **Reaction A1**, calculate the amount of sodium hydroxide present in 25.0 cm³ of **FA 1**.

[1]

- (vii) Calculate the concentration of sodium hydroxide, NaOH in **FA1**.

[1]

- (e) Give **one** reason why after 25.0 cm³ of **FA 1** is pipetted into a conical flask, the stock solution **FA 1** should be immediately covered with a sealant film.

State the effect of not covering **FA1** with a sealant film on the calculated concentration of NaOH.

.....

.....

.....[2]

[Total: 15]

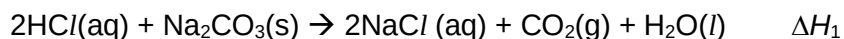
Question 2: To determine ΔH_4 , the enthalpy change of reaction between aqueous NaOH and CO₂ gas

The enthalpy change of reaction between aqueous NaOH and CO₂ gas as shown below cannot be found directly.



To determine this enthalpy change, several experiments need to be carried out.

You are to determine ΔH_1 , the enthalpy change of reaction between **FA3**, solid sodium carbonate and **FA2**, aqueous HCl.



You are provided with the following:

FA2, an aqueous solution of hydrochloric acid, HCl(aq)

FA3, solid sodium carbonate, Na₂CO₃(s)

Procedure

1. Using a 50 cm³ measuring cylinder, measure 25 cm³ of **FA2** into the styrofoam cup supported by a beaker.
2. Using a thermometer, measure and record the initial temperature (T_i) of the solution in the cup.
3. Weigh out accurately 2.50 g of **FA3** using a weighing bottle and record your weighings in an appropriate table.
4. Add the **FA3** into the styrofoam cup, and immediately cover the cup using the lid.
5. Insert the thermometer through the hole on the lid and stir the solution.
6. Measure and record the highest temperature (T_f) reached.
7. Calculate the temperature change, ΔT .
8. Reweigh the weighing bottle after emptying **FA3**. Record this mass.

(a) Table of results

Mass of empty weighing bottle /g	
Mass of weighing bottle and FA3 /g	
Mass of weighing bottle and residual FA3 /g	
Mass of FA3 used /g	2.504

[2]

Initial temperature of FA2 /°C	30.0
Final temperature/°C	36.0
Change in temperature/°C	+6.0

[4]

- (b) (i) Calculate the heat evolved in the reaction between **FA3** and **FA2**.
(specific heat capacity of the mixture is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)

[1]

- (ii) Determine the amount of Na_2CO_3 used. [A_r : Na: 23.0, C: 12.0, O: 16.0]

[1]

- (iii) Hence, calculate ΔH_1 , the enthalpy change of reaction between **FA3** and **FA2**.

[1]

- (c) **Determination of ΔH_2 , the enthalpy change of reaction between aqueous HCl and aqueous NaOH.**

(This experiment has been conducted for you)



Procedure

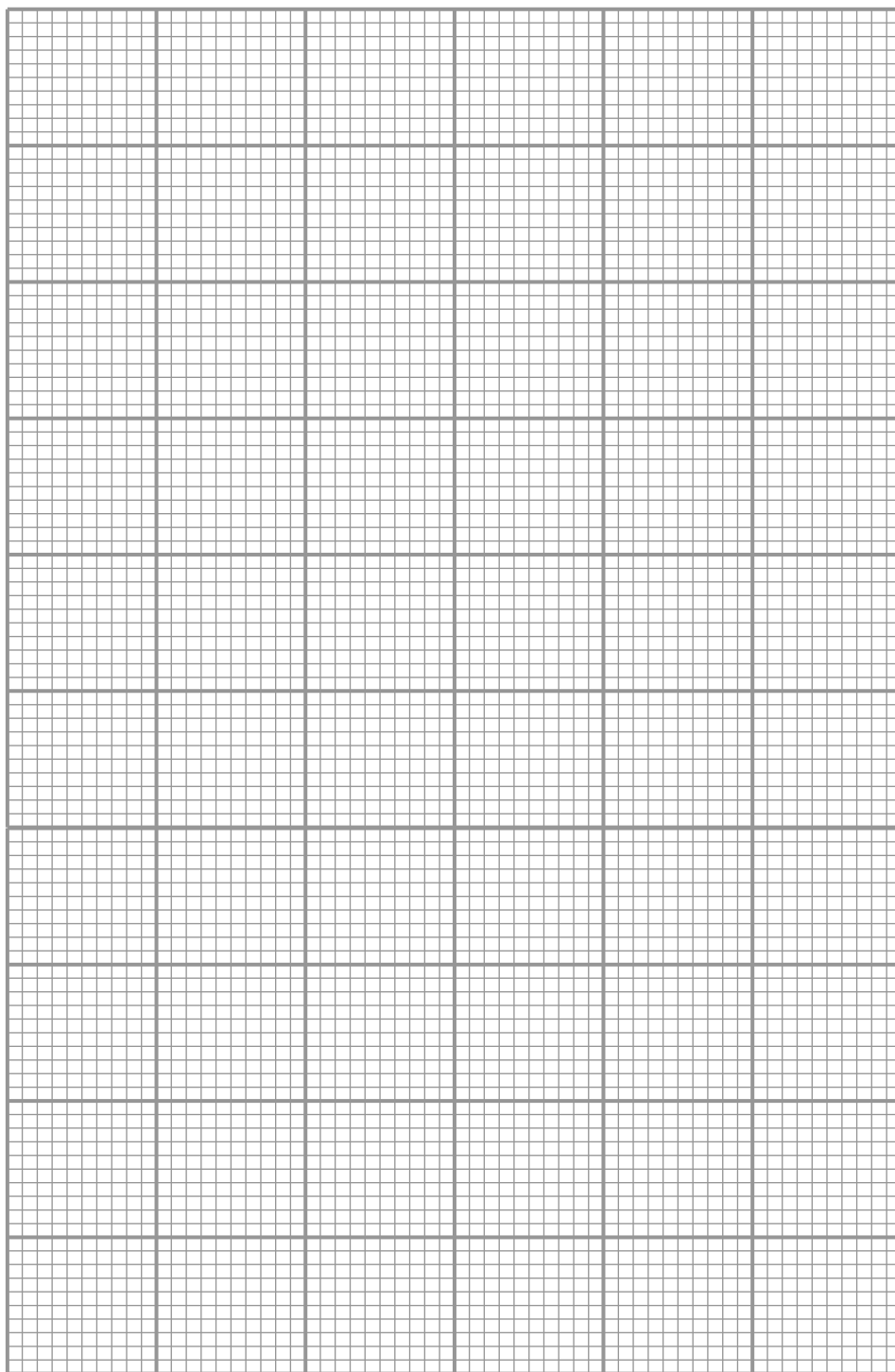
- Using the 50 cm^3 measuring cylinder, transfer 25 cm^3 of 2.5 mol dm^{-3} HCl into the polystyrene cup supported in a beaker.
- Stir and record the temperature of HCl every half minute for 2.5 minutes.
- At exactly 3 minutes, add 25 cm^3 of 2.0 mol dm^{-3} NaOH using a measuring cylinder to the HCl in the cup.
- Stir gently with the thermometer and continue to record the temperature every half minute from 3.5 minutes to 10 minutes.

The results from the experiment are given below.

Time/ min	Temperature/ $^{\circ}\text{C}$	Time/ min	Temperature/ $^{\circ}\text{C}$
0.0	25.2	5.5	37.8
0.5	25.2	6.0	37.6
1.0	25.2	6.5	37.4
1.5	25.2	7.0	37.2
2.0	25.2	7.5	37.0
2.5	25.2	8.0	36.8
3.0	-----	8.5	36.6
3.5	37.0	9.0	36.4
4.0	38.4	9.5	36.2
4.5	38.2	10.0	36.0
5.0	38.0		

Plot the graph of temperature against time using the data provided above. Use a scale of 1 large square to 2 min for the time axis and 1 large square to 2 °C for the temperature axis. The temperature axis need not start from zero.

Draw a best fit straight line for the points before the third minute. Draw a second best fit straight line for the points after the third minute. Extrapolate both lines to the third minute.



[2]

- (d) From your graph, read the minimum temperature, $T_{\min}$, and the maximum temperature, $T_{\max}$, at the third minute. Indicate these values clearly on your graph. Record these values in the space below and determine a value for the maximum temperature rise, ΔT , at the third minute.

$T_{\max} = \dots\dots\dots$

$T_{\min} = \dots\dots\dots$

$\Delta T = \dots\dots\dots$ [2]

- (e) (i) Calculate the heat evolved in the reaction between HCl and NaOH. (specific heat capacity of the mixture is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)

(ii) Hence, calculate ΔH_2 , the enthalpy change of reaction between HCl and NaOH. [1]

- (f) Aqueous NaOH is often used to remove CO_2 as from the air as shown below. [2]



Given that ΔH_3 , the enthalpy of solution of $\text{Na}_2\text{CO}_3(\text{s})$ in water is $-14.1 \text{ kJ mol}^{-1}$, use your answers for ΔH_1 and ΔH_2 to construct an energy cycle to determine ΔH_4 . You may use ΔH_1 as $-26.5 \text{ kJ mol}^{-1}$ and ΔH_2 as $-56.8 \text{ kJ mol}^{-1}$ if you did not manage to calculate these values in (b)(iii) and (e)(ii).

[4]

[Total: 18]

Question 3: Planning**Planning 1: Determination of the percentage of sodium carbonate in FA4.**

- (a) **FA4** contains a solid mixture of sodium carbonate and sodium hydroxide. You are to plan a gas collection experiment to determine the percentage by mass of sodium carbonate in **FA4**.

You may assume that the gas collected is at room temperature and pressure.

[Ar: Na: 23.0, C: 12.0, O: 16.0]

You are provided with the following:

- **FA2**: 100 cm³ of an aqueous solution of HCl.
 - solid **FA4**
 - a 100 cm³ graduated gas syringe
 - the equipment normally found in a school or college laboratory
- (i) Calculate an appropriate mass of **FA4** which you would use for the reaction by considering the capacity of the gas syringe and assuming that **FA4** is purely Na₂CO₃.

[2]

- (ii) Draw the experimental set-up.

[2]

Planning 2: Determination of the water of crystallisation in FA5, hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

(b) **FA5** contains hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$. You are to plan a gravimetry experiment to determine the value of x .

You are provided with the following:

- 1.00 g of **FA5**
- a boiling tube
- Bunsen burner
- the equipment normally found in a school or college laboratory

Outline, step by step, the procedure for the method you would use to

- carry out the weighing of a sample of **FA5** solid,
- remove the water of crystallisation (dehydration) in **FA5** by heating, and
- ensure that the dehydration is complete.

.....[3

[Total: 7]

END OF PAPER